MIL-P-36714B
6 October 1976
SUPERSEDING
MIL-P-36714A
18 April 1975
MIL-P-36714
13 October 1970 and
Amendment - 2
15 August 1974

MILITARY SPECIFICATION

PLUG, EAR, NOISE PROTECTION, TRIPLE FLANGE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers ear plugs intended for protection of personnel exposed to noises of high intensity.

1.2 Classification. Ear plugs shall be of one type, one grade and color coded according to size as follows:

Size	Color code
Regular (medium)	International Orange
Small	Green
Large	Blue

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-585	- Boxes, Wood, Wirebound.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-676	- Boxes, Set-Up.

MILITARY

MIL-STD-10547	- Liners, Case, and Sheet, Overwrap; Water-vapor-proof or Waterproof, Flexible.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq., Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, Code ATT, 2800 S. 20th St., Phila., PA 19101, by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-601 Rubber, Sampling and Testing.

MILITARY

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center (DPSC)

No. 20604 PLUG, EAR, NOISE PROTECTION, TRIPLE FLANGE.

(Copies of specifications, standards, and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

THE UNITED STATES PHARMACOPEIAL CONVENTION, INC.

The Pharmacopeia of the United States of America (USP)

(Application for copies shall be addressed to the Mack Publishing Company, Easton, Pennsylvania 18042).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D-2240 Indentation Hardness of Rubber and Plastics by Means of a Durometer.

(Application for copies should be addressed to American Society for Testing and Materials, 1903 Race Street, Philadelphia, Pa. 19103).

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NATIONAL FORMULARY

Light Mineral Oil

(Application for copies should be addressed to National Formulary, Washington, DC 20037).

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. Ear plugs shall be fabricated from non-toxic, non allergenic, sterilizable vulcanized silicone elastomer which shall be resistant to ear wax, alcohol, skin oils, and common detergents. The color shall be according to size as indicated in 1.2 and 3.6. The material shall be such that the finished ear plugs shall comply with the physical requirements contained in 3.3 and shall retain their noise attenuation properties and shape when subjected to a temperature range of -10 to 120 Centigrade (C).

3.2 Style, design and dimensions. The ear plugs shall be of the size as specified, and shall comfortably fit small, medium or large ear openings, as applicable. When properly placed in the ear openings, the ear plugs shall attenuate high noise levels to a comfortable level. Style, design, and dimensions shall be in accordance with Defense Personnel Support Center (DPSC) Drawing 20604.

3.3 Physical requirements of ear plugs. Ear plugs or test specimens (coupons) as specified in Section 4, when tested as described in the appropriate portions of Section 4, both before and after being subjected to the aging test described herein, shall conform to the following physical requirements:

3.3.1 Tensile strength, ultimate elongation, and hardness.

Tensile strength (minimum pounds per square inch (p.s.i.)	500
Ultimate elongation (minimum) percent	600
Hardness (Shore Durometer)	20±2

3.3.2 Resistance to cold. The ear plugs shall show no visible signs of cracks or other damage after exposure at -29 C ± 2 C that would render them unfit for the purpose intended.

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* 3.3.3 Resistance to volume change. Changes in volume shall be determined in accordance with 4.4.3. The percentage change in volume of the elastomeric material in alcohol, sodium hydroxide or oil shall not exceed 10 percent.

3.4 Health hazard. The finished ear plug shall be nontoxic, shall not constitute a health hazard to personnel when used as intended, and shall be such that when test material samples are subjected to the skin sensitivity patch test specified in 4.4.6, there shall be no evidence of inflammation or other adverse skin reactions caused by the test material on the human subjects used in the tests. The contractor shall provide for the performance of these tests by, or under the supervision of, a doctor of medicine.

3.5 Identification marking. Each ear plug shall be colored according to size and in addition each ear plug shall have a single raised letter present indicating its size as follows:

Size	Color
Medium (M)	International Orange
Small (S)	Green
Large (L)	Blue

3.6 Workmanship. The finished ear plugs shall conform to the highest standards of commercial molding practice. Ear plugs shall be free from defects which detract from their appearance or may impair their serviceability. They shall be free from objectionable odor at the time of delivery, and shall not develop an obnoxious odor during use and storage.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

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4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the elastomeric material, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the National stock number (NSN), the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

* 4.1.4 Skin sensitivity patch test. A certificate (or certificates) for each lot of ear plugs being delivered shall be forwarded to the procuring activity by the contractor to the effect that the ear plugs are nontoxic and meet the requirements of 3.4. Included in these certificates shall be records of the patch tests performed (see 4.4.6), listing the subjects tested by name, and the results of observations made every 24 hours on each subject. These records of tests and results shall be certified by the doctor of medicine who performed, or supervised performance of, the patch tests. In addition, each contractor's certificate shall include a statement that the samples used in the patch tests covered by the test records are from the same batch of elastomer (identical in composition and processing) as the ear plugs in the lot. In addition, when inspection is performed at source, a copy of the certificate (or certificates) shall be furnished to the cognizant Government inspector.

4.2 Sampling.

* 4.2.1 Sample specimens (coupons) for determining physical requirements of ear plugs. Sample specimens (coupons) for determining compliance of ear plugs with tests for tensile strength, ultimate elongation, hardness, resistance to volume change, skin sensitivity, and accelerated aging shall be prepared from 0.08 inch sheets of the elastomeric material used in the manufacture of the lots of ear plugs under test.

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4.2.2 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in table I. The unit of product shall be one ear plug.

TABLE I. Sampling criteria for examination.

Type of defect	Inspection level	AQL (percent defective) 1/
Major	S-3	1.0
Minor	S-2	2.5

1/If acceptance number is not exceeded, lot shall be acceptable. If rejection number is equaled or exceeded, inspection at level II may be performed to determine final acceptance or rejection.

4.2.3 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in table II.

TABLE II. Sampling criteria for tests.

Specimen	Characteristic	Requirements	Test procedure	Insp. level	AQL (percent defective)
Coupon	Tensile strength	3.3.1	4.4.1	S-1	1/
Coupon	Ultimate elongation	3.3.1	4.4.1	S-1	1/
Coupon	Hardness	3.3.1	4.4.2	S-2	1/
Plug, ear	Low temp. flexibility	3.3.2	4.4.4	S-1	1/
Coupon	Volume change	3.3.3	4.4.3.1 4.4.3.2 4.4.3.3	S-1	1/
Coupon	Skin sensitivity	3.4	4.4.6	S-2	1/
Coupon	Accelerated aging	3.3 & 3.3.1	4.4.5	S-1	1/
Plug, ear	Dimensions	3.2	-----	S-1	1.0

1/ Acceptance number shall be zero for all sample sizes.

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4.3 Examination. The ear plugs shall be examined to determine compliance with specification requirements. Examination shall be conducted in accordance with MIL-STD-105 and the following classification of defects. Examination shall include but shall not be limited to the defects listed in table III.

TABLE III. Classification of defects.

Categories	Defects
Major	
101	Presence of foreign matter or loose particles.
102	Lack of uniformity of color.
103	Presence of feathers, burrs, or unspecified protrusions which prevent proper seating in ear canal.
Minor	
201	Presence of feathers and burrs which do not interfere with function.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Tensile strength and ultimate elongation. Specimens shall be fabricated from sample specimens (coupons) specified in 4.2.1. The tests shall be conducted for tensile strength and ultimate elongation in accordance with FED-STD-601, Methods 4111 and 4121, respectively. Die No. III of Method 4111 shall be used for cutting the specimens.

4.4.2 Hardness. Sample specimens as specified in 4.2.1 shall be used. The specimens shall be superimposed until a stack, 1/4 inch high (minimum) results. The hardness shall be determined in accordance with ASTM Method D2240. At least 10 hardness determinations shall be made.

4.4.3 Resistance to volume change.

* 4.4.3.1 In ethyl alcohol. The weight of water displaced on a suitable balance having a sensitivity of 0.5 mg. by each of two, 1 by 2 by 0.08 inch specimens, (see 4.2.1) shall be determined before and after 24-hour immersion at 70° Fahrenheit (F.) + 2°F. in ethyl alcohol conforming to alcohol (ethanol) 92.3 to 93.8 percent by weight, of the current edition of the Pharmacopeia of the United States (USP). In determining the displacement after the 24-hour immersion, after removal, specimens shall be dipped in fresh ethyl alcohol, lightly blotted with filter paper, and the determination made within 5 minutes after removal from the ethyl alcohol. The displacement after the ethyl alcohol immersion, minus the initial displacement, divided by the initial displacement, multiplied by 100, shall be considered the percentage increase in volume for each specimen. The two results shall be averaged.

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4.4.3.2 In sodium hydroxide solution (NaOH). The percentage of increase in volume after 24-hour immersion at 70° F. ± 2° F. in a 0.1 percent (by weight) sodium hydroxide solution (prepared from sodium hydroxide conforming to requirements of the Pharmacopeia of the United States, and Purified Water, USP) shall be determined by the method described in 4.4.3.1.

4.4.3.3 In oil. The percentage of increase in volume after 24-hour immersion at 70° F. ± 2°F. in a light mineral oil, conforming to the requirements for Light Mineral Oil given in the National Formulary, shall be determined by the method described in 4.4.3.1.

4.4.4 Resistance to cold. Ear plugs shall be subjected to a temperature of -29°C. ± 2°C. for 3 hours. Immediately after removal from the cold chamber, they shall be manually compressed and flattened between two smooth wood surfaces under 25 pounds pressure (total load).

4.4.5 Aging test. Specimens (see 4.2.1) and ear plugs shall be given an accelerated aging by subjecting them to dry circulating air at a temperature of 158° F. ± 2° F. for 48 hours. Final determination of the properties of the aged specimens shall be made not less than 20 hours after removal from the oven.

4.4.6 Skin sensitivity patch test. One-half-inch squares (0.08-inch thick, see 4.2.1) of the material used in the fabrication of the ear plugs shall be applied to the skin of at least 10 human subjects (not less than one square applied to each subject) by means of adhesive tape and shall remain in place for 72 hours, except for removal to permit examination of the skin test area after every 24 hours (within that 72 hour period) for any evidence of inflammation, or other adverse skin reactions. If the material unduly irritates the skin, the test patch shall be removed immediately.

4.5 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing and marking comply with the requirements of Section 5. Defects shall be scored as specified in table IV. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of defects per hundred units.

TABLE IV. Examination of preparation for delivery.

Examine	Defect
Containers	Not as specified.
Contents	Not as specified.
Markings	Omitted; incorrect; illegible; improper size, location, sequence, or method of application
Packaging and packing materials.	Missing or damaged.
Workmanship	Bulging or distortion of containers.
	Cushioning inadequate, improper or missing.

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4.5.1 Examination of closure and reinforcement for containers. When shipping containers are required to comply with PPP-B-636, examination for defects in closure and reinforcement shall be in accordance with the appendix to that specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, or C, as specified (see 6.2).

5.1.2 Level A.

5.1.2.1 Unit package. Twenty four plugs of one size shall be wrapped in tissue paper and packaged in a snug fitting box constructed in accordance with PPP-B-566 or PPP-B-676 except that commercial colors are acceptable. Closure shall be secured to prevent spilling of contents under normal handling.

5.1.2.2 Intermediate package. Twelve unit packages containing plugs of one size only shall be packaged in a snug box constructed in accordance with PPP-B-566 or PPP-B-676 except that commercial colors are acceptable. Closure shall be secured to prevent spilling of contents under normal handling.

5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Forty eight intermediate packages containing plugs of one size only shall be packaged in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Fiberboard boxes shall conform to the special requirements as specified in PPP-B-636. Closure and strapping shall be as specified in the appendix of the applicable box specification. In addition, each wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed in accordance with the appendix of PPP-B-636.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

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5.2.2 Level B. Forty eight intermediate packages containing plugs of one size only shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure of wood boxes shall be as specified in the appendix of the applicable box specification. Closure of fiberboard boxes shall conform to method II of PPP-B-636. Fiberboard boxes shall conform to the special requirements specified in PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.2.4 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. This specification covers ear plugs which shall comfortably fit all average ear openings and serve to attenuate high noise levels to a comfortable level.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and the procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size (see 1.2 and 3.6).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(d) Applicable National Stock Number.

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6.3 This specification covers the following items appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
6515-00-442-4818	PLUG, EAR, Hearing Protection, Regular (Medium), International Orange, Triple Flange, 24's.
6515-00-442-4821	PLUG, EAR, Hearing Protection, Small, Green, Triple Flange, 24s.
6515-00-467-0092	PLUG, EAR, Hearing Protection, Large, Blue, Triple Flange, 24s.

6.4 Changes. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:
Army-MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6515-2003

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